

Heart Valves

Aortic calcification and stenosis (LPA, PALMD, TEX41, IL6, ALPL, FADS1/2), mitral regurgitation and prolapse (LMCD1, TNS1, BAG3, TBX5), bicuspid valve, polygenic scores and loci load

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. The strongest genetic factor of aortic stenosis is absent: Sérgio F has none of the alleles of Lp(a) high **LPA** (rs10455872 A/A, rs3798220 T/T, rs74617384 A/A). The polygenic score of Lp(a) is in percentile 75.0 (typical). The two signals of **IL6** in the valve are without risk alleles, and the same happens with **FADS1/2** (rs174554 G/G). In the mitral valve, the prolapse score is in percentile 54.8 (typical), the loci load in the 65.6 and the two strongest loci (LMCD1 and TNS1) are without risk alleles.

Unfavourable. The calcification of the aortic valve has a polygenic charge above average: the load of 492 loci of aortic stenosis is in percentile 95.1 and one of the two large scores is in the 88.8 (the other is in the 45.2, typical). They contribute to the homozygous genotypes of risk in TEX41, ALPL, HMGA2, FGF23 and SCARB1, 1 allele in **PALMD** and 1 in the locus of LDL **SORT1**. The genetic area of the aortic valve is in percentile 22.0 (below the median, but typical).

What does it mean? Without high genetic Lp(a), the risk of Sérgio F comes from the sum of many common loci linked to mineral metabolism (ALPL, FGF23), LDL (SORT1) and valvular development (PALMD, TEX41, HMGA2). This risk acts mainly after the 60–70 years and is modifiable by known factors: LDL and ApoB as low as possible, all life. The mitral valve has typical genetics.

Attachment to the Aorta report and aneurysms (§87). This report already shows, with the same values, the LPA (rs10455872 A/A, rs3798220 T/T), the PGS005252 aortic stenosis score (percentile 45.2), the mitral prolapse PGS002271 (percentile 54.8), the LMCD1 A/A, the TNS1 G/G and the TEX41 C/C. This report deepens what is not there: **PALMD**, the calcification pathway (ALPL, FGF23, HMGA2, SORT1), mitral regurgitation, rare genes **NOTCH1** and **DCHS1** and the aging point of view.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (Beagle 5.5, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Keyvariants** 28 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 19 of the chip, 9 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 0 SNPs needed conversion to the complementary strand and 6 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 6 PGS Catalog scores calculated locally with 4.325.583 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MAIS PROVÁVEL/ALTO; ≤ 20 : MENOS PROVÁVEL/BAIXO; between 20 and 80: TÍPICO.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 565 loci, 525 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In aortic stenosis, mitral prolapse and Lp(a), a higher percentile = more risk. In the aortic valve area, a higher percentile = larger valve (better).

Aortic stenosis

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Aortic stenosis	PGS005252 (Small 2025, LDpred2, 1.12 million variants)	1.086.572 1.119.377 (weighted coverage 96%)	45,2	TIPIC	typical
Aortic stenosis	PGS005254 (Kany 2025, PRS-CS, 1.11 million variants)	1.080.816 1.110.912 (weighted coverage 97%)	88,8	MAIS PROVABLE	unfavourable
Non-rheumatic aortic valve disease	PGS001025 (Tanigawa 2022, 36 variants)	26 36 (weighted coverage 85%)	47,7	TIPIC	typical
Aortic valve area	PGS005257 (Kany 2025, PRS-CS, 1.11 million variants)	1.080.959 1.111.632 (weighted coverage 97%)	22,0	TIPIC	typical

Mitral valve

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Prolapse of mitral valve	PGS002271 (Roselli 2022, PRS-CS, 1.10 million variants)	1.066.402 1.097.364 (weighted coverage 96%)	54,8	TIPIC	typical

Lipoprotein(a)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
Lipoprotein(a)	PGS000752 (Dron 2021, LDpred, 11 thousand variants)	10.808 11.446 (weighted coverage 90%)	75,0	TIPIC	typical

4. Key variants by topic

Variants with replicated effect on aortic valve calcification, mitral prolapse and bicuspid valve.

Lp(a) and aortic calcification

Lipoprotein(a) is the strongest genetic factor of calcified aortic stenosis: oxidized phospholipids that activate the calcification of interstitial cells of the valve.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LPA	rs10455872	LPA intron 25	A/A	chip	G (0 copy/s)	favourable	LPA intronic variant that marks small apo(a) and high Lp(a) isoforms; OR ~1.6–2 calcified aortic stenosis by allele (Thanassoulis 2013)	Strong
LPA	rs3798220	I4399M	T/T	chip	C (0 copy/s)	favourable	Ile4399Met (Ile1891Met) of apo(a): High Lp(a); coronary disease and aortic stenosis	Strong
LPA	rs74617384		A/A	chip	T (0 copy/s)	favourable	Main sign of LPA in the largest GWAS of aortic stenosis [GWAS Catalog: Aortic stenosis, allele T, OR 1.43, p=5E-261, PMID 41419686]	Strong

Calcification and inflammation of the aortic valve

Loci of the largest GWAS of aortic stenosis (Small 2025 and Kany 2025). Each common allele changes the risk in 5–15%; what counts is the sum. From the point of view of aging, calcified aortic stenosis is an active disease and not a simple wear: oxidized LDL and Lp(a) enter the leaflet, interstitial cells inflame, become senescent and undergo a phenotype of osteoblast, and lack of pyrophosphate (ALPL) and excess phosphate (FGF23) facilitate mineralization. The prevalence goes from ~0.2% to 50–59 years to ~10% after 80. PALMD (palmdelfine) maintains interstitial cells in the quiescent state; the risk allele reduces their expression.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
PALMD / LINC01708	rs6702619		T/G	chip	G (1 copy/s)	unfavourable (light)	Locus PALMD (1p21): less palmdelfine expression in the valve, more calcification (Thériault 2018; Helgadottir 2018 [GWAS Catalog: Aortic stenosis, allele G, OR 1.14, p=2E-124, PMID 41419686])	Strong
TEX41	rs2252641		C/C	chip	C (2 copy/s)	unfavourable	Locus TEX41 (2q22): aortic stenosis and bicuspid valve; valvular development [GWAS Catalog: Aortic stenosis, C allele, OR 1.10, p=4E-66, PMID 41419686]	Strong
IL6	rs1800797		G/G	chip	A (0 copy/s)	favourable	IL6 promoter (–597, in connection with –174): IL-6 signaling and inflammatory valve calcification [GWAS Catalog: Aortic stenosis, allele A, OR 1.11, p=4E-78, PMID 41419686]	Moderate
IL6	rs1474347		A/A	chip	C (0 copy/s)	favourable	Second sign of IL6 in aortic stenosis (valvular inflammation) [GWAS Catalog: Aortic stenosis, C allele, OR 1.11, p=2E-79, PMID 41419686]	Moderate

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ALPL	rs12141569		C/C	chip	C (2 copy/s)	unfavourable	Non-specific alkaline phosphatase of the tissue: hydrolyzes the pyrophosphate that stops the mineralization [GWAS Catalog: Aortic stenosis, C allele, OR 1.07, p=2E-38, PMID 41419686]	Moderate
FADS1 / FADS2	rs174554		G/G	chip	A (0 copy/s)	favourable	Fatty acid desaturations: more conversion to arachidonic acid (omega-6) and more procalcifying eicosanoids (Chen 2020) [GWAS Catalog: Aortic stenosis, allele A, OR 1.08, p=6E-46, PMID 41419686]	Moderate
CELSR2 / PSRC1 (SORT1)	rs602633		T/G	chip	G (1 copy/s)	unfavourable (light)	LDL locus (SORT1): more LDL throughout life, more aortic stenosis [GWAS Catalog: Aortic stenosis, allele G, OR 1.08, p=8E-31, PMID 41419686]	Strong
NAV1	rs650720		G/T	chip	T (1 copy/s)	unfavourable (light)	NAV1 locus (1q32) of aortic stenosis [GWAS Catalog: Aortic stenosis, allele T, OR 1.07, p=2E-35, PMID 41419686]	Moderate
HMGA2	rs56344960		C/C	imputation DR2=0.99	C (2 copy/s)	unfavourable	Locus HMGA2 (12q14): size/ area of the aortic valve and stenosis [GWAS Catalog: Aortic stenosis, C allele, OR 1.09, p=3E-33, PMID 41419686]	Moderate
FGF23	rs17773299		T/T	imputation DR2=0.95	T (2 copy/s)	unfavourable	Phosphorus-FGF23 axis: mineral metabolism and calcification [GWAS Catalog: Aortic stenosis, allele T, OR 1.10, p=7E-28, PMID 41419686]	Moderate
SCARB1	rs10846744		C/C	chip	C (2 copy/s)	unfavourable	SR-BI receptor of HDL: reverse cholesterol transport [GWAS Catalog: Aortic stenosis, C allele, OR 1.08, p=9E-27, PMID 41419686]	Moderate
ADAMTS7	rs4243085		C/C	chip	G (0 copy/s)	favourable	Matrix metalloproteinase; also coronal locus [GWAS Catalog: Aortic stenosis, allele G, OR 1.06, p=1E-21, PMID 41419686]	Moderate
SLC22A3	rs9295128		G/G	chip	G (2 copy/s)	typical	Locus SLC22A2/SLC22A3 (6q25, LPA neighbor): the common allele G (~99%) is associated with less stenosis; the rare allele A more	Moderate

Prolapse and mitral regurgitation

Mitral prolapse is a myxomatous degeneration (excess of proteoglycans and fragmentation of collagen and elastin) of the leaflets. Mitral regurgitation is the most frequent valvular disease after 65 years (~10% after 75): it results from prolapse or rupture of cords (primary form) or dilation of the auricle and ventricle (secondary form, linked to atrial fibrillation and heart failure). The genes TNS1, LMCD1, DCHS1 and FLNA act in the development of leaflets and in the signaling TGF- β ; DCHS1 (cadherin) and FLNA cause rare family forms that the chip does not read.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
LMCD1	rs171408		A/A	chip	G (0 copy/s)	favourable	LMCD1 locus (3p25) of mitral prolapse (Dina 2015) [GWAS Catalog: Mitral valve prolapse, allele G, OR 1.32, p=1E-11, PMID 26301497]	Moderate
TNS1	rs12465515		G/G	chip	C (0 copy/s)	favourable	Locus TNS1 (2q35): tensina-1, focal adhesion of valvular fibroblasts (Dina 2015) [GWAS Catalog: Mitral valve prolapse, allele C, OR 1.25, p=3E-11, PMID 26301497]	Moderate
ALPK3	rs35828350		A/G	imputation DR2=0.99	A (1 copy/s)	unfavourable (light)	Locus ZNF592/ALPK3 (15q25) of the mitral prolapse (Roselli 2022) [GWAS Catalog: Mitral valve prolapse, allele A, OR 1.21, p=2E-13, PMID 35245370]	Moderate
BAG3	rs17099139		C/G	imputation DR2=0.99	C (1 copy/s)	unfavourable (light)	BAG3: co-chaperone of cardiac cells; mitral prolapse and cardiomyopathy (Roselli 2022) [GWAS Catalog: Mitral valve prolapse, allele C, OR 1.19, p=3E-11, PMID 35245370]	Moderate
TBX5	rs2555005		A/C	imputation DR2=0.80	A (1 copy/s)	unfavourable (light)	TBX5: transcription factor of cardiac development [GWAS Catalog: Mitral valve prolapse, allele A, OR 1.15, p=1E-8, PMID 35245370]	Moderate
RBM20	rs12573386		C/C	imputation DR2=0.94	C (2 copy/s)	unfavourable	RBM20: splicing of titin; mitral prolapse [GWAS Catalog: Mitral valve prolapse, allele C, OR 1.16, p=5E-9, PMID 35245370]	Moderate
SMG6	rs112258894		T/T	imputation DR2=1.00	T (2 copy/s)	unfavourable	SMG6: mitral prolapse with surgery [GWAS Catalog: Mitral valve after mitral valve emergence, allele T, OR 1.30, p=2E-11, PMID 35245370]	Moderate
LTBP2	rs888414		G/G	imputation DR2=0.91	G (2 copy/s)	unfavourable	LTBP2: extracellular matrix and TGF-β [GWAS Catalog: Mitral valve after mitral valve emergence, allele G, OR 1.26, p=9E-10, PMID 35245370]	Moderate
SPTBN1	rs12713274		A/A	imputation DR2=1.00	A (2 copy/s)	unfavourable	SPTBN1: spectrin; TGF-β [GWAS Catalog: Mitral valve prolapse, allele A, OR 1.18, p=3E-9, PMID 35245370]	Moderate
ATXN2	rs10774625		G/G	chip	G (2 copy/s)	unfavourable	Locus ATXN2/SH2B3 (12q24): pleiotropic [GWAS Catalog: Mitral valve prolapse, allele G, OR 1.15, p=4E-9, PMID 35245370]	Weak

Bicuspid valve (congenital)

The bicuspid aortic valve is congenital (1-2% of the population) and is the main cause of aortic stenosis before 65-70 years. The rare variants of NOTCH1 (via Notch, which stops the osteogenic differentiation of valvular cells) cause familial bicuspid and early calcification; they are not in the chip. The common genes only give a probability; the echocardiogram confirms or excludes.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
MUC4	rs2246901		A/A	chip	C (0 copy/s)	favourable	Aortic valve MUC4 [GWAS Catalog: Bicuspid aortic valve, C allele, OR 1.23, p=2E-9, PMID 35727948]	Weak
LINC01708 (PALMD)	rs11166276		C/T	chip	T (1 copy/s)	unfavourable (light)	Locus PALMD also associated with the bicuspid valve [GWAS Catalog: Bicuspid aortic valve, allele T, OR 1.21, p=1E-18, PMID 41645906]	Moderate

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of aortic stenosis, mitral prolapse and valvular disease, with alleles oriented towards more risk, compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Aortic stenosis	529 (492)	514 de 984	489	95,1	unfavourable
Prolapse of mitral valve	22 (20)	18 de 40	20	65,6	typical
Vascular disease (general)	14 (13)	11 de 26	13	46,2	typical

6. Favourable

- **LPA without high Lp(a) alleles** (rs10455872 A/A, rs3798220 T/T, rs74617384 A/A): the largest genetic factor of aortic stenosis (OR ~1.5–2 by allele) is absent. Lp(a) score in percentile 75.0. **Strong.**
- **IL6 without risk alleles** in both signs of aortic stenosis (rs1800797 G/G, rs1474347 A/A): less inflammatory calcification via IL-6. **Moderate.**
- **FADS1/2 rs174554 G/G**: without the allele attached to more arachidonic acid and more stenosis. **Moderate.**
- **ADAMTS7** without risk allele; **SLC22A3** with common genotype. **Moderate.**
- **Typical mitral prolapse**: score in percentile 54.8, loci load in 65.6; LMCD1 and TNS1 (the most effective loci) without risk alleles. **Moderate.**
- **Bicuspid valve**: MUC4 without risk allele; general valvular disease (loci) in percentile 46.2. **Weak.**

7. Unfavourable

- **Polygenic load of high aortic stenosis:** 492 loci in percentile 95.1; Kany 2025 score in the 88.8 (Small 2025 in 45.2 and 36 variants in 47.7). **Moderate.**
- **Risk homozygosity in TEX41, ALPL, HMGA2, FGF23 and SCARB1:** are common alleles with small effect (OR 1.07–1.10 each), but add up. ALPL and FGF23 point to the mineralization pathway (pyrophosphate and phosphate). **Moderate.**
- **PALMD rs6702619 T/G** and **SORT1 rs602633 T/G** (1 allele each): less valvular palmdelfin and more LDL throughout life. Strong for loci.
- **Aortic valve area** in percentile 22.0 (below the median, still typical). **Weak.**
- **Mitral prolapse:** homozygosity in the common alleles of RBM20, SMG6, LTBP2, SPTBN1 and ATXN2 and 1 allele in ALPK3, BAG3 and TBX5, with no effect on the global score (typical). **Weak.**

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Remodelling of the extracellular matrix	Aortic calcification (loci in percentile 95.1) and mitral myxomatous degeneration (typical)	Unfavourable (mild)
Stiffness of the Extracellular Matrix	ALPL and FGF23 (mineralization) and HMGA2: stiffer leaflets with age	Unfavourable
Vascular Aging	Without high genetic Lp(a); LDL (SORT1) with 1 allele	Mixed
Chronic Inflammation	IL6 without risk alleles in the valve	Favourable
Cell Senescence	Senescent interstitial valvular cells pass to osteoblasts; loci PALMD/TEX41	Unfavourable (mild)
Metabolic Homeostasis Decline	Axis fosfato-FGF23 (FGF23 T/T) and LDL throughout life	Unfavourable (mild)

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Baseline echocardiogram** (valves, aorta root and exclusion of bicuspid valve), and cardiac auscultation in routine consultations; repeat each 3–5 years from the 60, or earlier if there is breath, lack of effort, chest pain or dizziness. *Suggestion (not applied)*.
2. **LDL and ApoB as low as possible, all life**: the accumulated LDL is causal in the initial calcification of the valve (the SORT1 and the stenosis score point to this direction). Keep the current strategy of LDL/ApoB low and the fiber at most. *Suggestion (not applied)*.
3. **Lp(a): measure once in life**, if not yet measured. The genes predict a typical low value, but the analysis confirms (LPA has copy number variants that the chip does not read). *Suggestion (not applied)*.
4. **Mineralization**: avoid high-dose calcium supplements without indication and the excess of additives phosphates (cola refrigerants, processed meats, melted cheeses); vitamin D is at the dose of the plan, without megadoses. Vitamin K2 has not been shown to stop aortic stenosis in trials, so it is not suggested for this purpose. *Suggestion (not applied)*.
5. **Omega-3 (3.2-4 g/day, already in the plane)**: compensates for the FADS pathway (EPA/DHA preformed); there is no evidence that arrests the stenosis, but it is consistent with the plaque and inflammation. *Suggestion (not applied)*.
6. **Oral health**: hygiene and regular consultations; reduces bacteriemia, which matters if there is a valve altered or prosthesis. *Suggestion (not applied)*.
7. **Aerobic and regular strength training**, zero tobacco and zero alcohol (already fulfill): protect the valve and the ventricle. If stenosis arises, review with the cardiologist the limits of high intensity training. *Suggestion (not applied)*.

10. Limitations

- The chip does not read rare variants of great effect: **NOTCH1** (bicuspid valve and early calcification), **DCHS1**, **FLNA** and **DZIP1** (family mitral prolapse), or the number of kringle IV-2 copies of LPA.
- The two large scores of aortic stenosis disagree (45.2 and 88.8 percentiles); the loci load agrees with the highest.
- Aortic stenosis loci have small effects and add SNPs in linkage imbalance; the load is descriptive.
- The presence of bicuspid valve is only confirmed by echocardiogram.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local imputation (Beagle 5.5, public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The published allele of effect is compared with the alleles of the Imposed VCF; in the palindromics the strand + (standard of the GWAS Catalog and the PGS Catalog) is assumed.
- **Polygenic scores:** Harmonized GRCh38 files of the PGS Catalog (6 scores: aortic stenosis, aortic valve disease, valve area, mitral prolapse and Lp(a)). Score = Δ coefficient $\times$ dosage. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog ($p \leq 510^{-8}$), alleles guided by the published effect signal, empirical calibration in the same reference population.
- **Interpretation:** reference literature (Thanassoulis 2013 for LPA; Thériault 2018 and Helgadottir 2018 for PALMD and TEX41; Small 2025 and Kany 2025 for aortic stenosis; Dina 2015 and Roselli 2022 for mitral prolapse), GWAS Catalog and ClinVar. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Lp(a) and aortic calcification	LPA	rs10455872	A/A	chip	G	0	strand +	without the risk allele
Lp(a) and aortic calcification	LPA	rs3798220	T/T	chip	C	0	strand +	without the risk allele
Lp(a) and aortic calcification	LPA	rs74617384	A/A	chip	T	0	strand + (palindromic; assumed +)	without the risk allele
Calcification and inflammation of the aortic valve	PALMD / LINC01708	rs6702619	T/G	chip	G	1	strand +	1 copy of risk allele
Calcification and inflammation of the aortic valve	TEX41	rs2252641	C/C	chip	C	2	strand +	2 risk allele copies
Calcification and inflammation of the aortic valve	IL6	rs1800797	G/G	chip	A	0	strand +	without the risk allele
Calcification and inflammation of the aortic valve	IL6	rs1474347	A/A	chip	C	0	strand +	without the risk allele
Calcification and inflammation of the aortic valve	ALPL	rs12141569	C/C	chip	C	2	strand +	2 risk allele copies
Calcification and inflammation of the aortic valve	FADS1 / FADS2	rs174554	G/G	chip	A	0	strand +	without the risk allele
Calcification and inflammation of the aortic valve	CELSR2 / PSRC1 (SORT1)	rs602633	T/G	chip	G	1	strand +	1 copy of risk allele
Calcification and inflammation of the aortic valve	NAV1	rs650720	G/T	chip	T	1	strand +	1 copy of risk allele
Calcification and inflammation of the aortic valve	HMGA2	rs56344960	C/C	imputation DR2=0.99	C	2	strand +	2 risk allele copies
Calcification and inflammation of the aortic valve	FGF23	rs17773299	T/T	imputation DR2=0.95	T	2	strand +	2 risk allele copies
Calcification and inflammation of the aortic valve	SCARB1	rs10846744	C/C	chip	C	2	strand + (palindromic; assumed +)	2 risk allele copies
Calcification and inflammation of the aortic valve	ADAMTS7	rs4243085	C/C	chip	G	0	strand + (palindromic; assumed +)	without the risk allele
Calcification and inflammation of the aortic valve	SLC22A3	rs9295128	G/G	chip	G	2	strand +	common genotype (without the rare allele of risk)
Prolapse and mitral regurgitation	LMCD1	rs171408	A/A	chip	G	0	strand +	without the risk allele
Prolapse and mitral regurgitation	TNS1	rs12465515	G/G	chip	C	0	strand + (palindromic; assumed +)	without the risk allele
Prolapse and mitral regurgitation	ALPK3	rs35828350	A/G	imputation DR2=0.99	A	1	strand +	1 copy of risk allele

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Prolapse and mitral regurgitation	BAG3	rs17099139	C/G	imputation DR2=0.99	C	1	strand + (palindromic; assumed +)	1 copy of risk allele
Prolapse and mitral regurgitation	TBX5	rs2555005	A/C	imputation DR2=0.80	A	1	strand +	1 copy of risk allele
Prolapse and mitral regurgitation	RBM20	rs12573386	C/C	imputation DR2=0.94	C	2	strand + (palindromic; assumed +)	2 risk allele copies
Prolapse and mitral regurgitation	SMG6	rs112258894	T/T	imputation DR2=1.00	T	2	strand +	2 risk allele copies
Prolapse and mitral regurgitation	LTBP2	rs888414	G/G	imputation DR2=0.91	G	2	strand +	2 risk allele copies
Prolapse and mitral regurgitation	SPTBN1	rs12713274	A/A	imputation DR2=1.00	A	2	strand +	2 risk allele copies
Prolapse and mitral regurgitation	ATXN2	rs10774625	G/G	chip	G	2	strand +	2 risk allele copies
Bicuspid valve (congenital)	MUC4	rs2246901	A/A	chip	C	0	strand +	without the risk allele
Bicuspid valve (congenital)	LINC01708 (PALMD)	rs11166276	C/T	chip	T	1	strand +	1 copy of risk allele